

Every Separable Metric Space Extends to an Extreme One

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Abstract

The bounded-by-1 pseudometrics on a set X form a compact convex set $\bar{\mathcal{M}}(X)$. For finite X this is a rational polytope, so its extreme points have rational distances and satisfy the rigidity constraints that come with lying on many facets. We show that no such restriction survives passage to infinite X : every separable bounded-by-1 pseudometric space (X, d) is the restriction of an extreme point of $\bar{\mathcal{M}}(\tilde{X})$ for some $\tilde{X} \supseteq X$ with $\tilde{X} \setminus X$ countable. The local structure of an extreme point may therefore be arbitrary, irrational distances included. The proof glues finite extreme metrics of prescribed rational distances, supplied by the bowtie construction of the companion paper, onto a countable dense subset, after subdividing each irrational distance into a summable sequence of rational ones.

The extension admits a canonical form, functorial in isometric embeddings of countable spaces, and the original space is recoverable from it as the derived set of its completion. It follows that graph isomorphism Borel-reduces to isometry of extreme Polish metric spaces: extreme spaces admit no classification by real-number invariants, nor by countable structures. A structural obstruction confines this method below the universal orbit equivalence relation, and the resulting gap is posed as a question.

1 Introduction

Let X be a set and let $\bar{\mathcal{M}}(X)$ denote the set of bounded-by-1 pseudometrics on X : symmetric functions $d: X \times X \rightarrow [0, 1]$ vanishing on the diagonal and satisfying the triangle inequality. Under pointwise operations $\bar{\mathcal{M}}(X)$ is convex, and in the product topology it is compact, being a closed subset of $[0, 1]^{X \times X}$. Write $\text{ex}(\bar{\mathcal{M}}(X))$ for its extreme points.

For finite X with $|X| = n$, $\bar{\mathcal{M}}(X)$ is a polytope in $\mathbb{R}^{\binom{n}{2}}$ cut out by the triangle inequalities and the bounds $0 \leq d \leq 1$, all of which have integer coefficients. Its extreme points are therefore rational, and each is pinned by $\binom{n}{2}$ independent tight constraints—a strong rigidity condition. Which finite metrics are extreme is a delicate question with a substantial literature [1, 4, 9], and even the classification of the extreme rays of the associated cone is complete only for at most six points.

The situation for infinite X is the opposite, and that is the first content of this paper.

Theorem 1.1. *Let (X, d) be a separable bounded-by-1 pseudometric space. Then there is a set $\tilde{X} \supseteq X$ with $\tilde{X} \setminus X$ countable, and an extreme point $\tilde{d} \in \text{ex}(\bar{\mathcal{M}}(\tilde{X}))$, with $\tilde{d}|_{X \times X} = d$. The space $(\tilde{X}, \tilde{d})$ is again separable.*

So extremality imposes no constraint whatever on the isometry type of a separable bounded-by-1 pseudometric space, up to the addition of countably many points: the class of extreme separable bounded-by-1 pseudometric spaces is universal for the class of all of them. In particular:

Corollary 1.2. *For countable X , an extreme point of $\bar{\mathcal{M}}(X)$ may take irrational values, and indeed may realize any prescribed countable subset of $[0, 1]$ among its distances.*

The mechanism is a trade between two kinds of rigidity. A perturbation of a pseudometric must vanish wherever a distance is 0 or 1, and must respect every degenerate triangle; in a finite space these constraints compete with the $\binom{n}{2}$ degrees of freedom and are hard to satisfy in bulk. Given countably many new points one may instead *install* the constraints: every distance one wishes to freeze is embedded in a finite extreme metric that realizes it, and the finitely many new distances that the embedding creates are either equal to 1, hence frozen outright, or forced by a degenerate triangle through the gluing locus, hence not free either. Irrational distances, which no finite extreme metric can realize, are first broken into a summable sequence of rational ones.

The second content concerns classification, and begins from an observation of Dmitri Nikshych: Theorem 1.1 reads as a *nonclassification* result, the world of extreme separable metric spaces being at least as rich as the world of all of them. We make this precise. The extension admits a canonical form E , functorial in isometric embeddings of countable metric spaces (Theorem 6.3), which in particular resolves the dependence of the construction on its many choices; and E is reversible: the input is recoverable from the output as the derived set of its completion (Theorem 7.4). Composing the two yields a Borel reduction of graph isomorphism to isometry of extreme Polish metric spaces (Theorem 7.5), so extreme Polish metric spaces cannot be classified by real-number invariants, nor by countable structures (Corollary 7.6). Against the Gao–Kechris theorem [5], which places isometry of all Polish metric spaces at the maximal complexity among orbit equivalence relations of Polish group actions, a gap remains; we prove that no construction canonical merely at the level of a marked countable dense set can close it (Proposition 7.7), and pose the sharpened question (Question 7.8).

Section 2 records what a perturbation of a pseudometric can do; the bound $|\varepsilon| \leq \min\{d, 1 - d\}$ of Lemma 2.3 is used throughout and is what makes extremality a property detectable on a dense subset (Lemma 2.5). Section 3 proves the gluing lemma, Section 4 the subdivision lemma, and Section 5 assembles them. Section 6 constructs the canonical extension and Section 7 the recovery and the reduction. Section 8 collects the remaining questions.

We use one input from the finite theory, proved in Section 5 as Proposition 5.1: every rational number in $(0, 1]$ occurs as a distance in some extreme point of $\bar{\mathcal{M}}(F)$ with F finite. It rests on the generalized bowtie metrics of the companion paper [7], and is the only place the finite classification enters.

The results of Sections 2–5 have been formalized in Lean 4 against Mathlib; formalization of Sections 6–7 is in progress. Section 9 records the correspondence, the scope, and the deposit.

2 Perturbations

Throughout, a *metric* means a bounded-by-1 pseudometric, and (X, d) ranges over such spaces of arbitrary cardinality. We write $d(x, y)$ and speak of the *distance* of the pair $\{x, y\}$.

Definition 2.1. A *perturbation* of $d \in \bar{\mathcal{M}}(X)$ is a symmetric function $\varepsilon: X \times X \rightarrow \mathbb{R}$ vanishing on the diagonal such that both $d + \varepsilon$ and $d - \varepsilon$ lie in $\bar{\mathcal{M}}(X)$.

Perturbations detect extremality, by the same argument as in a finite-dimensional polytope.

Proposition 2.2. *$d \in \text{ex}(\bar{\mathcal{M}}(X))$ if and only if the only perturbation of d is $\varepsilon = 0$.*

Proof. If $\varepsilon \neq 0$ is a perturbation then $d = \frac{1}{2}(d + \varepsilon) + \frac{1}{2}(d - \varepsilon)$ exhibits d as the midpoint of a segment with distinct endpoints in $\bar{\mathcal{M}}(X)$, so d is not extreme. Conversely if $d = \frac{1}{2}(a + b)$ with $a \neq b$ in $\bar{\mathcal{M}}(X)$, then $\varepsilon = \frac{1}{2}(a - b)$ is a nonzero perturbation. $\square$

The next two lemmas are elementary and do all of the analytic work in this paper.

Lemma 2.3. *Let ε be a perturbation of $d \in \bar{\mathcal{M}}(X)$. Then*

$$|\varepsilon(x, y)| \leq \min\{d(x, y), 1 - d(x, y)\} \quad \text{for all } x, y \in X.$$

In particular $\varepsilon(x, y) = 0$ whenever $d(x, y) \in \{0, 1\}$.

Proof. Both $d + \varepsilon$ and $d - \varepsilon$ take values in $[0, 1]$. Nonnegativity of $d - \varepsilon$ and of $d + \varepsilon$ gives $|\varepsilon| \leq d$; the bounds $d \pm \varepsilon \leq 1$ give $|\varepsilon| \leq 1 - d$. $\square$

Lemma 2.4. *Let ε be a perturbation of $d \in \bar{\mathcal{M}}(X)$ and suppose $d(x, z) = d(x, y) + d(y, z)$. Then*

$$\varepsilon(x, z) = \varepsilon(x, y) + \varepsilon(y, z).$$

Proof. Applying the triangle inequality to $d + \varepsilon$ at $[x, y, z]$ and subtracting the hypothesis gives $\varepsilon(x, z) \leq \varepsilon(x, y) + \varepsilon(y, z)$; the same computation for $d - \varepsilon$ gives the reverse inequality. $\square$

Lemma 2.3 has a consequence with no finite-dimensional analogue: a perturbation is continuous with respect to d , whether or not $d + \varepsilon$ has any a priori relation to the topology of d . Extremality is therefore visible on any dense subset.

Lemma 2.5. *Let $d \in \bar{\mathcal{M}}(X)$, let $A \subseteq X$ be dense in the pseudometric topology of d , and let ε be a perturbation of d with $\varepsilon|_{A \times A} = 0$. Then $\varepsilon = 0$.*

Proof. Write $\rho = d + \varepsilon \in \bar{\mathcal{M}}(X)$. Fix $x, y \in X$ and choose sequences $a_k \rightarrow x$ and $b_k \rightarrow y$ in A . By Lemma 2.3, $\rho(a_k, x) \leq d(a_k, x) + |\varepsilon(a_k, x)| \leq 2d(a_k, x) \rightarrow 0$, and likewise $\rho(b_k, y) \rightarrow 0$. The triangle inequality for ρ gives

$$|\rho(a_k, b_k) - \rho(x, y)| \leq \rho(a_k, x) + \rho(b_k, y) \rightarrow 0,$$

so $\rho(a_k, b_k) \rightarrow \rho(x, y)$. But $\varepsilon(a_k, b_k) = 0$, so $\rho(a_k, b_k) = d(a_k, b_k)$, and $d(a_k, b_k) \rightarrow d(x, y)$ by the same estimate applied to d . Hence $\rho(x, y) = d(x, y)$, that is $\varepsilon(x, y) = 0$. $\square$

Corollary 2.6. *If $Y \subseteq X$ and ε is a perturbation of $d \in \bar{\mathcal{M}}(X)$, then $\varepsilon|_{Y \times Y}$ is a perturbation of $d|_{Y \times Y}$. Consequently, if $d|_{Y \times Y}$ is an extreme point of $\bar{\mathcal{M}}(Y)$ then every perturbation of d vanishes on $Y \times Y$.*

Proof. The restriction of a bounded-by-1 pseudometric to a subset is one, and restriction is compatible with the operations $d \pm \varepsilon$. $\square$

3 Gluing along a finite set

Two metric spaces meeting in a finite set on which they agree may be amalgamated, and the amalgam can be chosen so that every new distance is either maximal or forced.

Lemma 3.1. *Let (X, d) and (Y, ρ) be bounded-by-1 pseudometric spaces with $F := X \cap Y$ finite and non-empty, and with $d|_{F \times F} = \rho|_{F \times F}$. Define $\omega: (X \cup Y)^2 \rightarrow [0, 1]$ by $\omega|_{X \times X} = d$, $\omega|_{Y \times Y} = \rho$, and*

$$\omega(x, y) := \min\left\{\min_{z \in F}(d(x, z) + \rho(z, y)), 1\right\} \quad (x \in X \setminus Y, y \in Y \setminus X). \quad (1)$$

Then $\omega \in \bar{\mathcal{M}}(X \cup Y)$, and for every $x \in X \setminus Y$ and $y \in Y \setminus X$ either $\omega(x, y) = 1$, or there is $z \in F$ with

$$\omega(x, y) = \omega(x, z) + \omega(z, y).$$

Proof. Symmetry, vanishing on the diagonal and the range $[0, 1]$ are clear from (1); the final dichotomy holds because F is finite, so the inner minimum is attained, and $\omega(x, z) = d(x, z)$, $\omega(z, y) = \rho(z, y)$ for $z \in F$.

For the triangle inequality write $P = X \setminus Y$, $Q = Y \setminus X$. A triple with all three points in X , or all three in Y , is governed by d or by ρ . Since $F \subseteq X$ and $F \subseteq Y$, the remaining triples have at least one point in P and at least one in Q , leaving three cases up to the symmetry exchanging (X, d, P) with (Y, ρ, Q) .

Case 1: $x \in P$, $y \in Q$, $z \in F$. First, $\omega(x, y) \leq d(x, z) + \rho(z, y) = \omega(x, z) + \omega(z, y)$ directly from (1). Next, $\omega(x, z) \leq \omega(x, y) + \omega(y, z)$: if $\omega(x, y) = 1$ this reads $d(x, z) \leq 1 + \rho(y, z)$, which holds as $d \leq 1$; otherwise $\omega(x, y) = d(x, w) + \rho(w, y)$ for some $w \in F$, and

$$d(x, z) \leq d(x, w) + d(w, z) = d(x, w) + \rho(w, z) \leq d(x, w) + \rho(w, y) + \rho(y, z),$$

using $d|_F = \rho|_F$. The inequality $\omega(z, y) \leq \omega(x, y) + \omega(x, z)$ follows by exchanging the roles of x and y .

Case 2: $x, x' \in P$, $y \in Q$. For $\omega(x, y) \leq \omega(x, x') + \omega(x', y)$: if $\omega(x', y) = 1$ the right side is at least $1 \geq \omega(x, y)$; otherwise $\omega(x', y) = d(x', w) + \rho(w, y)$ with $w \in F$, and (1) gives

$$\omega(x, y) \leq d(x, w) + \rho(w, y) \leq d(x, x') + d(x', w) + \rho(w, y) = \omega(x, x') + \omega(x', y).$$

The inequality $\omega(x', y) \leq \omega(x, x') + \omega(x, y)$ is the same statement with x and x' exchanged. For $\omega(x, x') \leq \omega(x, y) + \omega(x', y)$: if either term on the right equals 1 the inequality holds since $d \leq 1$; otherwise $\omega(x, y) = d(x, w) + \rho(w, y)$ and $\omega(x', y) = d(x', w') + \rho(w', y)$ with $w, w' \in F$, whence

$$d(x, x') \leq d(x, w) + d(w, w') + d(w', x') \leq d(x, w) + \rho(w, y) + \rho(y, w') + d(w', x'),$$

again using $d|_F = \rho|_F$ and the triangle inequality for ρ at $[w, y, w']$. The right side is $\omega(x, y) + \omega(x', y)$.

Case 3: $x \in P$, $y, y' \in Q$. Symmetric to Case 2. □

Remark 3.2. The truncation at 1 in (1) is what keeps ω inside $\bar{\mathcal{M}}$; without it one obtains the usual amalgamated metric, the pushout of $(X, d) \leftarrow (F, d|_F) \rightarrow (Y, \rho)$ in the category of pseudometric spaces and 1-Lipschitz maps. The dichotomy in Lemma 3.1 says that each new distance sits at the ceiling or on a geodesic through F , and it is this dichotomy, rather than the amalgamation itself, that the proof of Theorem 1.1 consumes.

4 Subdividing a distance

No finite extreme metric has an irrational distance, so an irrational distance cannot be frozen by the method of Section 5 directly. It can be written as a sum of rational distances along a countable geodesic, after which freezing the summands freezes the sum.

Lemma 4.1. *Let (X, d) be a bounded-by-1 pseudometric space and let $x_0, y_0 \in X$ with $t := d(x_0, y_0) > 0$. There exist a countable set $C = \{c_1, c_2, \dots\}$ disjoint from X and a metric $\omega \in \bar{\mathcal{M}}(X \cup C)$ with $\omega|_{X \times X} = d$ such that, writing $c_0 := x_0$:*

1. $\omega(c_i, c_{i+1}) = a_i$ for a sequence of positive rationals with $\sum_{i \geq 0} a_i = t$;
2. $\omega(c_0, c_j) = \sum_{i < j} a_i$ and $\omega(c_j, y_0) = t - \sum_{i < j} a_i$ for all $j \geq 1$;

3. every perturbation ε of ω satisfies $\varepsilon(x_0, y_0) = \sum_{i \geq 0} \varepsilon(c_i, c_{i+1})$, the series converging absolutely.

Proof. Choose positive rationals $a_0, a_1, \dots$ with $\sum_{i \geq 0} a_i = t$, put $s_0 = 0$ and $s_j = \sum_{i < j} a_i$, so $s_j \uparrow t$ and $0 < a_i < t \leq 1$. Let $S = \{s_0, s_1, \dots\} \cup \{t\} \subseteq [0, 1]$ carry the metric inherited from $\mathbb{R}$, and let C^+ be a copy of S in which the points s_0 and t are identified with x_0 and y_0 respectively and the points s_j with $j \geq 1$ are new; write c_j for the point at s_j and $C = \{c_1, c_2, \dots\}$. The metric inherited from $\mathbb{R}$ agrees with d on $C^+ \cap X = \{x_0, y_0\}$, since $|t - 0| = t = d(x_0, y_0)$, and that intersection is finite, so Lemma 3.1 supplies $\omega \in \mathcal{M}(X \cup C)$ restricting to d on X and to the real-line metric on C^+ . Items (1) and (2) are then immediate.

For (3), let ε be a perturbation of ω . By (2) the triples $[c_0, c_j, c_{j+1}]$ and $[c_0, c_j, y_0]$ are degenerate for ω , so Lemma 2.4 gives

$$\varepsilon(c_0, c_{j+1}) = \varepsilon(c_0, c_j) + \varepsilon(c_j, c_{j+1}), \quad \varepsilon(c_0, c_j) + \varepsilon(c_j, y_0) = \varepsilon(x_0, y_0),$$

whence $\varepsilon(c_0, c_j) = \sum_{i < j} \varepsilon(c_i, c_{i+1})$ for every j . By Lemma 2.3, $|\varepsilon(c_i, c_{i+1})| \leq a_i$, so the series converges absolutely, and $|\varepsilon(c_j, y_0)| \leq \omega(c_j, y_0) = t - s_j \rightarrow 0$. Letting $j \rightarrow \infty$ in the second display gives $\varepsilon(x_0, y_0) = \sum_{i \geq 0} \varepsilon(c_i, c_{i+1})$. $\square$

5 Proof of the theorem

The finite theory enters exactly once.

Proposition 5.1. *For every rational $q \in (0, 1]$ there are a finite set F , an extreme point $\rho \in \text{ex}(\mathcal{M}(F))$, and points $u, v \in F$ with $\rho(u, v) = q$.*

Proof. Write $q = a/b$ with $b \geq 2$ and $1 \leq a \leq b$; this is possible for every rational in $(0, 1]$, doubling numerator and denominator when the reduced denominator is 1.

Let B_b denote the generalized bowtie metric of [7] on the cells $P_1, \dots, P_{b+1}$ of sizes $1, 2, 2, \dots, 2$, so that F has $2b + 1$ points. Its interior cells $P_2, \dots, P_b$ all have size 2, and there are $b + 1 \geq 3$ cells, so the side condition on two-cell bowties is vacuous; and $2b + 1 \geq 5$. The generalized bowtie theorem of [7] therefore makes B_b a generator of an extreme ray of the metric cone on F . Its values on distinct points are the level differences $1, \dots, b$ together with the cell-mate value 2, so $\frac{1}{b}B_b$ takes exactly the values $\frac{1}{b}, \frac{2}{b}, \dots, \frac{b}{b}$ and has maximum 1. A generator of an extreme ray with maximum value 1 is an extreme point of the body [7], and a/b occurs among its values. $\square$

Remark 5.2. The single family B_b covers every rational in $(0, 1]$, where the corresponding corollary of [7] splits into the cases $q = 2$ and $q \geq 3$ with different cell patterns.

The formalization of Proposition 5.1 does not pass through the metric cone at all, and the argument is worth recording, since the final step is shorter in the body than in the cone. The tight triples of B_b are exactly the *chains*, on points of three strictly increasing levels, and the *vees*, on two cell-mates and a point of an adjacent level. Let ε be a perturbation of $\frac{1}{b}B_b$. For a four-cycle x, u, y, v whose four sides join consecutive cells and whose two diagonals are realized as vees through both of the remaining two points, Lemma 2.4 applied to the four vees gives

$$\varepsilon(x, u) + \varepsilon(u, y) = \varepsilon(x, v) + \varepsilon(v, y), \quad \varepsilon(x, u) + \varepsilon(x, v) = \varepsilon(u, y) + \varepsilon(v, y),$$

whence $\varepsilon(u, y) = \varepsilon(x, v)$ and $\varepsilon(x, u) = \varepsilon(v, y)$. Propagating these equalities across and between blocks gives a common value λ of ε on all pairs joining consecutive cells, and then Lemma 2.4 on vees and chains gives $\varepsilon = \lambda B_b$ throughout. The apex P_1 and any point of P_{b+1} are at distance 1, so Lemma 2.3 forces ε to vanish there, and that value is $\lambda b/b = \lambda$. Hence $\lambda = 0$ and $\varepsilon = 0$.

Proof of Theorem 1.1. Fix a countable dense $A \subseteq X$.

The construction. We build an increasing chain $X = X_0 \subseteq X_1 \subseteq X_2 \subseteq \dots$ of sets and a compatible chain of metrics $d = \omega_0, \omega_1, \dots$ with $\omega_{k+1}|_{X_k \times X_k} = \omega_k$, each step being one of:

- (R) *Rigidify a rational pair.* Given $u, v \in X_k$ with $\omega_k(u, v) = q \in \mathbb{Q} \cap (0, 1)$, take F and $\rho \in \text{ex}(\bar{\mathcal{M}}(F))$ from Proposition 5.1 realizing q at a pair of points of F , identify those two points of F with u and v , and let $X_{k+1} = X_k \cup F$ with ω_{k+1} the metric of Lemma 3.1 for the intersection $\{u, v\}$.
- (S) *Subdivide an irrational pair.* Given $u, v \in X_k$ with $\omega_k(u, v) \notin \mathbb{Q}$, let $X_{k+1} = X_k \cup C$ and $\omega_{k+1} = \omega$ as furnished by Lemma 4.1 applied to (X_k, ω_k) and the pair u, v .

Maintain a task list, initially the pairs $\{u, v\} \subseteq A$ with $0 < d(u, v) < 1$. A task with rational distance is discharged by a step of type (R); a task with irrational distance is discharged by a step of type (S), which appends to the list the countably many new pairs $\{c_i, c_{i+1}\}$, all of rational distance $a_i \in (0, 1)$. Each step adds at most countably many points and at most countably many tasks, so dovetailing arranges that every task is discharged at some finite stage. Put $\tilde{X} = \bigcup_k X_k$ and let $\tilde{d}$ be the common extension of the ω_k ; this is well defined because any two points of $\tilde{X}$ lie in a common X_k and the ω_k are compatible. Then $\tilde{d} \in \bar{\mathcal{M}}(\tilde{X})$, $\tilde{d}|_{X \times X} = d$, and $\tilde{X} \setminus X$ is a countable union of countable sets, hence countable. Since $A \cup (\tilde{X} \setminus X)$ is countable and dense in $\tilde{X}$, the extension is separable.

Every perturbation vanishes on $A \times A$. Let ε be a perturbation of $\tilde{d}$. Let $u, v \in A$ be distinct. If $d(u, v) \in \{0, 1\}$ then $\varepsilon(u, v) = 0$ by Lemma 2.3. Otherwise $\{u, v\}$ was a task.

If $d(u, v)$ is rational, the task was discharged by a step of type (R), which placed u, v inside a copy of a finite F carrying an extreme ρ . By Corollary 2.6, ε vanishes on $F \times F$, so $\varepsilon(u, v) = 0$.

If $d(u, v)$ is irrational, the task was discharged by a step of type (S), and Lemma 4.1(3) gives $\varepsilon(u, v) = \sum_i \varepsilon(c_i, c_{i+1})$. Each pair $\{c_i, c_{i+1}\}$ was itself appended to the task list with rational distance $a_i \in (0, 1)$ and so was later discharged by a step of type (R); by the previous paragraph $\varepsilon(c_i, c_{i+1}) = 0$ for every i , and hence $\varepsilon(u, v) = 0$.

Every perturbation vanishes on $X_k \times X_k$, by induction on k . For $k = 0$: $\varepsilon|_{X \times X}$ is a perturbation of d by Corollary 2.6, it vanishes on $A \times A$ by the previous paragraph, and A is dense in X , so Lemma 2.5 gives $\varepsilon|_{X \times X} = 0$.

Assume ε vanishes on $X_k \times X_k$ and let $X_{k+1} = X_k \cup N$ be the next step, glued along the finite set $F_0 = X_k \cap N$. A pair with both points in X_k is covered by the inductive hypothesis. For a pair with both points in N : if the step was of type (R), N carries a finite extreme metric and Corollary 2.6 applies; if the step was of type (S), N is the chain $\{c_j\}$ together with $F_0 = \{u, v\}$, every distance $\omega_{k+1}(c_i, c_j)$ with $i < j$ is the sum $\sum_{i \leq l < j} a_l$ of consecutive ones by Lemma 4.1(2), so repeated use of Lemma 2.4 expresses $\varepsilon(c_i, c_j)$ as $\sum_{i \leq l < j} \varepsilon(c_l, c_{l+1})$, which vanishes as above; the pairs $\{c_j, v\}$ are handled by the second display in the proof of Lemma 4.1(3). Finally let $x \in X_k \setminus N$ and $y \in N \setminus X_k$. By Lemma 3.1 either $\omega_{k+1}(x, y) = 1$, in which case $\varepsilon(x, y) = 0$ by Lemma 2.3, or there is $z \in F_0$ with $\omega_{k+1}(x, y) = \omega_{k+1}(x, z) + \omega_{k+1}(z, y)$, in which case Lemma 2.4 gives $\varepsilon(x, y) = \varepsilon(x, z) + \varepsilon(z, y)$; here $x, z \in X_k$ and $y, z \in N$, so both summands vanish by what precedes. This completes the induction.

Every pair of points of $\tilde{X}$ lies in some X_k , so $\varepsilon = 0$. By Proposition 2.2, $\tilde{d}$ is extreme. □

Proof of Corollary 1.2. Let $V \subseteq [0, 1]$ be countable. Realize V as the set of distances of some countable bounded-by-1 pseudometric space—for instance, take a countable set $\{p_v\}_{v \in V}$ together with a base point p_* , and define $d(p_*, p_v) = v$ and $d(p_v, p_w) = \max\{v, w\}$ for $v \neq w$, an ultrametric.

Apply Theorem 1.1; the extension restricts to this space, so its distances include V . The resulting space is countable, being countable plus countable. $\square$

6 The canonical extension

The construction of Section 5 depends on an enumeration of tasks, on the rationals chosen to subdivide each irrational distance, and on the finite extreme metric chosen to freeze each rational one. All three choices can be made canonically, and the enumeration can be eliminated outright, by decorating *ordered pairs* in two simultaneous stages.

Definition 6.1 (canonical selectors). For irrational $t \in (0, 1)$ let $t = \sum_{k \in K(t)} 2^{-k}$ be the binary expansion, $K(t)$ infinite, and let the *canonical subdivision* of t be the sequence of gaps $(2^{-k})_{k \in K(t)}$ in increasing order of k . For rational $q \in (0, 1)$ let $b(q)$ be the denominator of q in lowest terms, so $b(q) \geq 2$, and let the *canonical piece* for q be the scaled bowtie $\frac{1}{b(q)}B_{b(q)}$ of Proposition 5.1, with its designated ordered pair (apex, point of level a), where $q = a/b(q)$.

Definition 6.2 (canonical extension). Let (A, d) be a countable metric space. *Stage 1*: for every ordered pair $(u, v) \in A^2$ with $d(u, v)$ irrational, adjoin the subdivision chain from u to v with the canonical gaps, gluing the whole family at once. *Stage 2*: for every ordered pair (x, y) of points of the stage-1 space whose distance is a rational in $(0, 1)$ and which is either a pair from A or an adjacent gap pair of a stage-1 chain, adjoin the canonical piece for that distance with x at the apex, again gluing the family at once. Write $E(A)$ for the result.

Since each ordered pair is decorated, and (u, v) and (v, u) receive mirror-image decorations, no orientation is chosen; since the decorations are indexed by the pairs themselves, no enumeration is chosen; and the selectors of Definition 6.1 are functions of the distance alone. The proof of Theorem 1.1 applies verbatim—the task list there is exactly the index set here—so $E(A)$ is an extreme metric space extending A by countably many points.

Theorem 6.3 (functoriality). *E is a functor from countable bounded-by-1 pseudometric spaces with isometric embeddings to extreme ones: an isometric embedding $\psi: A \rightarrow B$ extends canonically to an isometric embedding $E(\psi): E(A) \rightarrow E(B)$ restricting to ψ , and $E(\psi' \circ \psi) = E(\psi') \circ E(\psi)$. In particular isometric countable spaces have isometric canonical extensions.*

Proof. ψ carries an ordered pair of A to an ordered pair of B at the same distance, hence with the same canonical subdivision or the same canonical piece; define $E(\psi)$ as ψ on A and as the identity on decoration patterns over matched pairs. Distances inside A are preserved by hypothesis; distances inside a decoration depend only on the distance of its pair; and a distance between two different decorations, or between a decoration and a base point, is by (1) a truncated minimum of sums of within-piece distances and base distances over the finitely many anchors of the pieces involved, every term of which is matched. The extra decorations present in $E(B)$ over unmatched pairs do not enter these formulas. Composition holds because patterns map by the identity. $\square$

Remark 6.4. Functoriality and minimality pull apart: a functor must decorate every pair uniformly, so it cannot fix even the extreme inputs with a distance in $(0, 1)$, while a minimal extension of an extreme space is the space itself. The canonical extension is visibly non-minimal. Whether minimal extreme extensions exist at all remains open; see Question 8.2.

7 Recovery and nonclassification

Theorem 1.1 says that extremality excludes no separable isometry type, and in this sense already reads as a nonclassification statement. This section makes the reading precise, in the framework of Borel reducibility [5, 6]: writing $\leq_B$ for Borel reducibility, $\cong_{\text{gi}}$ for graph isomorphism on countable graphs, and $\cong_i$ for isometry of Polish metric spaces, we place isometry of *extreme* Polish metric spaces above $\cong_{\text{gi}}$.

Extremality passes to completions, so the discussion may be conducted among Polish spaces.

Lemma 7.1. *Let $d \in \bar{\mathcal{M}}(X)$ be extreme and let $\bar{X}$ be the completion of (X, d) , with metric $\bar{d}$. Then $\bar{d}$ is an extreme point of $\bar{\mathcal{M}}(\bar{X})$.*

Proof. A perturbation of $\bar{d}$ restricts to a perturbation of d by Corollary 2.6, hence vanishes on $X \times X$; and X is dense in $\bar{X}$, so the perturbation vanishes by Lemma 2.5. $\square$

Extremality is moreover a Borel condition in the standard presentation of Polish metric spaces as completions of metrics on $\mathbb{N}$: the set $\bar{\mathcal{M}}(\mathbb{N})$ is a compact convex metrizable subset of $[0, 1]^{\mathbb{N} \times \mathbb{N}}$, its extreme points form a G_δ subset, and by Lemma 7.1 together with Corollary 2.6 a metric on $\mathbb{N}$ is extreme exactly when the metric of its completion is. Write $\cong_i^{\text{ext}}$ for the restriction of $\cong_i$ to this G_δ class.

The decorations of the canonical extension are metrically inert, while its chains accumulate exactly at the core; this makes the core definable from the isometry type.

Lemma 7.2 (isolation). *Every decoration point of $E(A)$ is an isolated point of $\overline{E(A)}$.*

Proof. Let p be a decoration point, belonging to the piece P glued along the finite anchor set F_P . By (1), the distance from p to any point outside P is at least $\delta_1 := \min_{z \in F_P} d(p, z)$, a minimum over a finite set of positive numbers. Within P : if P is a bowtie, P is finite and $\delta_2 :=$ its least positive distance to p is positive; if P is a chain and $p = c_j$, the within-piece distances from p are $|s - s_j|$ over the other partial sums s and the two endpoint distances s_j and $t - s_j$, all at least $\delta_2 := \min\{a_{j-1}, a_j, s_j, t - s_j\} > 0$. So no point of $E(A)$ other than p lies within $\min\{\delta_1, \delta_2\}$ of p , and the same holds in the completion. $\square$

Lemma 7.3. *Every limit point of $\overline{E(A)}$ lies in $\bar{A}$.*

Proof. Let $p_n \rightarrow z$ with the p_n distinct. If infinitely many p_n lie in A then $z \in \bar{A}$ at once, so suppose all are decoration points. If some piece contains infinitely many p_n , then z lies in the closure of that piece, which is the piece itself when a bowtie, and the piece together with its two endpoints when a chain; a chain has no limit points but its endpoints, so z is an endpoint, hence a base point of its stage. If instead the pieces are eventually distinct, then since for p, p' in distinct pieces (1) gives $d(p, p') \geq d(p, F_P)$, the Cauchy property forces $d(p_n, F_{P_n}) \rightarrow 0$ along a subsequence; choosing anchors $u_n \in F_{P_n}$ with $d(p_n, u_n) \rightarrow 0$ gives $u_n \rightarrow z$ with the u_n base points of the relevant stage. In either case z is a limit of stage-2 base points, that is, of points of A together with chain points; applying the same argument to a sequence of chain points, whose anchors lie in A , puts $z \in \bar{A}$, which is closed in $\overline{E(A)}$. $\square$

Theorem 7.4 (recovery). *Suppose every point of A has some point of A at irrational distance. Then the derived set of $E(A)$ is exactly $\bar{A}$. Consequently, for countable A and B satisfying the hypothesis,*

$$\overline{E(A)} \cong \overline{E(B)} \implies \bar{A} \cong \bar{B}.$$

Proof. By Lemma 7.3 the derived set is contained in $\bar{A}$. Conversely let $a \in A$ and pick $a' \in A$ with $d(a, a')$ irrational; the canonical chain for the ordered pair (a', a) has its partial sums converging to $d(a', a)$, so its points converge to a , and a is a limit point; a general point of $\bar{A}$ is a limit of points of A , and the derived set is closed. The consequence holds because an isometry of completions carries derived sets to derived sets. $\square$

The hypothesis is essential only for the containment $\bar{A} \subseteq \text{derived set}$; Lemmas 7.2 and 7.3 are unconditional.

Theorem 7.5. $\cong_{\text{gi}} \leq_B \cong_i^{\text{ext}}$.

Proof. Fix irrationals $\alpha = 1/\sqrt{3}$ and $\beta = 1/\sqrt{5}$, both in $(\frac{1}{3}, \frac{2}{3})$. For a graph G on $\mathbb{N}$ let X_G be $\mathbb{N}$ with $d(m, n) = \alpha$ for edges and β for non-edges, $m \neq n$. Any two values in $(\frac{1}{3}, \frac{2}{3})$ satisfy all triangle inequalities, so X_G is a metric space; it is complete and uniformly discrete, so $\bar{X}_G = X_G$; every point has points at irrational distance; and since $\alpha \neq \beta$, a bijection $\mathbb{N} \rightarrow \mathbb{N}$ is an isometry $X_G \rightarrow X_H$ if and only if it is an isomorphism $G \rightarrow H$.

The reduction is $G \mapsto$ a code of $\overline{E(X_G)}$: the carrier of $E(X_G)$ is $\mathbb{N}$ together with countably many canonically indexed decoration points, enumerated by a fixed bijection with $\mathbb{N}$, and each distance is given by an explicit formula—a within-piece value, or a truncated minimum over finitely many anchor routes—so the map is Borel. The output is extreme by Section 6 and Lemma 7.1.

If $G \cong H$ then $X_G \cong X_H$, so $\overline{E(X_G)} \cong \overline{E(X_H)}$ by Theorem 6.3 and Lemma 7.1. If conversely $\overline{E(X_G)} \cong \overline{E(X_H)}$, then $X_G \cong X_H$ by Theorem 7.4, hence $G \cong H$. $\square$

Corollary 7.6. *Isometry of extreme Polish metric spaces is not smooth, and is at least as complicated as the isomorphism relation of any class of countable structures. In particular extreme Polish metric spaces admit no Borel classification by real-number invariants, nor by countable structures.*

Proof. $\cong_{\text{gi}}$ is universal among isomorphism relations of classes of countable structures and is not smooth [6]; Theorem 7.5 puts $\cong_i^{\text{ext}}$ above it, and $\cong_i^{\text{ext}} \leq_B \cong_i$ by restriction. $\square$

By the theorem of Gao and Kechris [5], $\cong_i$ is Borel bireducible with the universal orbit equivalence relation induced by Borel actions of Polish groups, so $\cong_i^{\text{ext}}$ now sits between $\cong_{\text{gi}}$ and that maximum. It is tempting to close the gap by feeding E richer inputs. That route is closed, for a structural reason worth recording. Call a map F from codes to metric spaces *marked-canonical* if the isometry type of $F(A)$ depends only on the isometry type of A as a countable metric space—as E does by Theorem 6.3—rather than only on the completion $\bar{A}$.

Proposition 7.7. *No marked-canonical F with a recovery property—*isometry of outputs implies isometry of the countable inputs*—can Borel-reduce a universal orbit equivalence relation to isometry of its outputs: any equivalence relation reducible through such a pair of implications Borel-reduces to $\cong_{\text{gi}}$.*

Proof. The two implications together give $z E z' \iff A_z \cong A_{z'}$, isometry of countable metric spaces. A countable metric space is a countable structure in the countable language $\{R_q : q \in \mathbb{Q}\}$, $R_q(x, y) \iff d(x, y) < q$, with isomorphism equal to isometry, so this relation Borel-reduces to $\cong_{\text{gi}}$. A universal orbit equivalence relation does not: by the theory of turbulence [6] there are orbit equivalence relations not classifiable by countable structures, and these reduce to the universal one. $\square$

So the dependence of the extension on its countable dense set is not bookkeeping to be routed around; it is the exact boundary of the method, and Theorem 7.5 is optimal for every construction of this shape.

Question 7.8. Is $\cong_i^{\text{ext}}$ Borel bireducible with the universal orbit equivalence relation of Polish group actions—equivalently, is $\cong_i \leq_B \cong_i^{\text{ext}}$? By Proposition 7.7, an affirmative proof must produce extreme extensions whose isometry type depends only on the completion of the input, or avoid extension constructions altogether. A negative answer would be equally striking: extremality would then be a genuine simplifying constraint, placing $\cong_i^{\text{ext}}$ strictly between the countable-structure maximum and the universal one, a region where few natural equivalence relations are known to live.

8 Remaining questions

Question 8.1. Theorem 1.1 spends countability twice: once on the dense set A , and once on the subdivision of each irrational distance. For a bounded-by-1 pseudometric space of density character κ , does there exist an extreme extension $\tilde{X}$ with $|\tilde{X} \setminus X| \leq \kappa$?

The first use of countability appears inessential—Lemma 2.5 holds for a dense set of any cardinality, and the task list may be indexed by any well-ordering—while the second is genuine: a single irrational distance requires countably many points to subdivide, so a space with κ irrational distances calls for $\kappa \cdot \aleph_0$ new points, which equals κ for infinite κ . The obstruction to a transfinite recursion is bookkeeping rather than geometry, but the dovetailing argument above does not transcribe directly.

Question 8.2. Does every separable bounded-by-1 pseudometric space admit a *minimal* extreme extension, one no proper subspace of which containing X carries an extreme metric restricting to d ? Theorem 6.3 answers the functorial half of the canonicity question; by Remark 6.4 no functorial extension is minimal, so the two notions of canonicity diverge and the minimal one remains open.

A related point deserves recording. For finite X the extreme points of $\bar{\mathcal{M}}(X)$ are sparse and rigid, and the difficulty of the subject lies in identifying them. Theorem 1.1 shows that for countable X they are instead dense in a strong sense: every separable bounded-by-1 pseudometric space sits inside one. The quantitative content has migrated entirely into the finite theory, through Proposition 5.1—which distances a finite extreme metric can realize—and that question remains the interesting one.

9 Formal verification

The results of Sections 2–6, and of Section 7 apart from the exceptions marked below, have been formalized in Lean 4 against Mathlib. The development is in [3]; it compiles free of `sorry`, and each theorem named below depends only on `propext`, `Classical.choice` and `Quot.sound`. The formalization of Section 7 constructs no completions: isolation, limits and recovery are proved on the countable carriers, with isolation carrying an explicit radius, and Theorem 7.5’s equivalence is proved there for simple graphs, which is the generality its statement intends. Two entries are marked accordingly: [†]Lemma 7.1 is the one numbered result left in prose—its two-line proof consumes only Corollary 2.6 and Lemma 2.5, both formalized—and [‡]the containment of limit points in $\bar{A}$ is formalized in the direction the reduction uses. The descriptive-set-theoretic inputs rest on cited literature outside the scope of formalization, as marked.

Extremality is formalized twice: as the vanishing of every perturbation, and as membership in Mathlib’s `Set.extremePoints` for the convex set of bounded-by-1 pseudometrics, the two being identified by Proposition 2.2. Theorem 1.1 and Corollary 1.2 carry no hypotheses beyond those stated here.

Result	Status	Lean name
Theorem 1.1	formalized	<code>separable_extends_to_extreme'</code> <code>separable_extends_to_extremePoint'</code>
Corollary 1.2	formalized	<code>exists_countable_extreme_realizing'</code>
Definition 2.1	formalized	<code>IsPerturbation</code>
Proposition 2.2	formalized	<code>rigid_iff_extremePoint</code>
Lemma 2.3	formalized	<code>abs_pert_le</code>
Lemma 2.4	formalized	<code>pert_add_of_tight</code>
Lemma 2.5	formalized	<code>pert_eq_zero_of_dense</code>
Corollary 2.6	formalized	<code>pert_eq_zero_on_of_rigid_restrict</code>
Lemma 3.1	formalized	<code>glue_isBddPseudo</code> , <code>glue_dichotomy</code>
Remark 3.2	formalized	<code>glue_no_truncation_not_isBddPseudo</code>
Lemma 4.1	formalized	<code>pert_eq_tsum_of_geodesic</code>
Proposition 5.1	formalized	<code>exists_finite_rigid_realizing</code>
Remark 5.2	formalized	<code>Bowtie.bowtie_rigid</code>
§5 induction	formalized	<code>rigid_of_chain</code>
Theorem 6.3	formalized	<code>E_functor</code> , <code>E_functor_surj</code>
Lemma 7.1	prose only [†]	—
Lemmas 7.2, 7.3	formalized [‡]	<code>decor_isolated</code> , <code>core_is_limit</code>
Theorem 7.4	formalized, carrier form	<code>recovery</code>
Theorem 7.5, metric content	formalized	<code>main_equivalence</code>
Theorem 7.5, Borel wrapper	cites [5, 6]	—
Corollary 7.6, Prop. 7.7	cite [5, 6]	—
Questions 8.1, 8.2, 7.8	open questions	—

Proposition 5.1 is formalized by the direct argument of Remark 5.2 rather than by the route through the metric cone given in its proof above, so the machine-checked development depends on no result outside this paper.

Four statements were recorded in the development as deliberately false, to guard hypotheses that the proofs consume: the amalgam of Lemma 3.1 without its truncation at 1; the dichotomy of Lemma 3.1 with an infinite gluing locus; the sufficiency of a single round of gluing, without the subdivision of Lemma 4.1; and the conclusion of Lemma 2.5 without the hypothesis that ε is a perturbation. Each is refuted in the development by an explicit counterexample.

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